

Quarry V14: Receipt-Bound Deterministic Control for Reliable LLM Tool Use

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Abstract

We present Quarry V14, a content-blind reliability layer around a configurable Claude Sonnet 5 agent for CAR-bench Track 1. Instead of asking a second model to judge every action, V14 reconstructs a typed episode ledger from the visible conversation, advertised tool schemas, and call-ID-matched tool results. It checks proposed calls before dispatch, permits at most one same-model correction when a visible invariant fails, validates the correction again, and emits a causal audit receipt. A separate call-ID-bound postcondition completes a required multi-stop navigation disclosure without another LLM call. On a frozen 25-task public-validation panel with three trials per task, V14 obtained 14/25 strict Pass³, compared with 12/25 for the stock Sonnet 5 scaffold and 15/25 for our earlier V7-E3 controller. Five correction calls accounted for 1.23% of participant calls and 2.47% of participant inference cost, with 11.40 seconds median observed correction latency. V7-E3’s best raw score is reported rather than obscured, but its task-level gains occurred on non-intervention turns. V14 therefore claims a deployable pattern for sparse, state-grounded intervention whose scope, cost, postconditions, and failures can be independently audited, rather than universal causal uplift.

1 Motivation and Contribution

CAR-bench evaluates tool-using automotive assistants under incomplete capabilities, ambiguous requests, and domain policies [Kirmayr et al., 2026]. These settings expose a gap between plausible language and reliable execution: an agent can select an unavailable tool, reuse an identifier from the wrong namespace, claim success after a failed call, or omit a policy-mandated disclosure. Prompting alone leaves these failures stochastic and difficult to attribute.

Existing safeguards include model-based guard agents that translate policies into executable checks [Xiang et

al., 2025], runtime rule languages for intercepting agent actions [Wang et al., 2026], and provenance-based pre-dispatch alignment checks [She et al., 2026]. V14 does not claim runtime enforcement itself as new. Its contribution is an A2A-native combination of call-ID binding across proposals and results, payload-preserving pass-through, one-shot correction with strict re-validation, response postconditions, and re-derivable receipts under a stochastic harness.

Quarry V14 treats reliability as a control-plane problem. The underlying model still plans and speaks; a deterministic layer checks only invariants that can be derived from the current visible episode. The design has four goals: (1) preserve the stock event payload when no rule fires; (2) intervene sparsely and with bounded inference; (3) never use task identity, rewards, answer keys, or evaluator internals; and (4) bind every intervention to the proposal, visible evidence, dispatched action, and resulting tool output.

The main contribution is a transparent architecture that combines a typed runtime ledger, pre-dispatch guards, one bounded correction, strict post-validation, and response-level postconditions. The resulting receipts make an important distinction possible: raw Pass³ movement is useful for rank selection, whereas a mechanism claim requires an observed intervention and a complete evidence chain.

2 Method and Harness Architecture

2.1 Model and A2A Harness

The submitted agent exposes the official JSON-RPC A2A interface [A2A Protocol Working Group, 2026] and wraps the official Track 1 single-model scaffold. The participant model is configured by environment variable and was evaluated as anthropic/claude-sonnet-5 [Anthropic, 2026] through locked LiteLLM 1.85.1 [LiteLLM Contributors, 2026] and the direct Anthropic API. Thinking was disabled. The container runs as a non-root user on linux/amd64. Final responses preserve the benchmark’s turn_metrics, including prompt, completion, and thinking tokens, cost, model, call count, and mean LLM-call time; V14 adds measured controller latency.

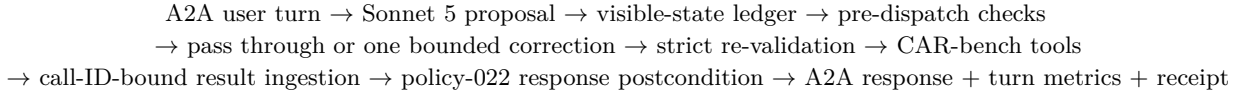


Figure 1: Quarry V14’s runtime path. Only visible A2A messages, advertised schemas, and current-episode tool results enter the controller. The deterministic footer path adds no model call.

2.2 Visible-State Ledger

For each A2A context, V14 rebuilds state from scratch rather than trusting model claims. User messages contribute explicit requests and confirmations. Assistant tool proposals create a mapping from tool-call IDs to tool names. A tool result updates state only when its call ID matches a pending proposal and its payload explicitly reports success. The ledger records grounded identifier namespaces, navigation state and waypoints, route metadata, successful and failed actions, and observed confirmations. Missing or malformed evidence does not become a fact.

This reconstruction is deliberately content-blind with respect to evaluation: the source contains no CAR-bench task IDs or family selectors, and the runtime does not receive rewards or task-level metrics. It sees exactly the information available to an ordinary participant agent.

2.3 Sparse Pre-Dispatch Control

Before a proposal reaches the environment, typed predicates check malformed calls or JSON arguments, unavailable tools, missing required parameters, identifier-namespace mismatches, contradictory navigation state, and missing confirmation. Valid sibling calls are preserved. Where a correction is mechanical, grounded identifiers can be repaired or an advertised prerequisite can be inserted. Otherwise, the controller makes at most one additional call to the same model with a compact diagnostic containing only failed invariants and visible ledger facts.

Every corrected proposal passes a stricter second validator. This stage also rejects schema violations, ungrounded identifiers, unadvertised parameters, and duplicate call IDs. Remaining invalid calls are withheld; the agent does not iterate against evaluator feedback. Diagnosis failures preserve the stock path, while required intervention receipts fail loudly when an audit sink is configured.

2.4 Call-ID-Bound Response Postcondition

CAR policy 022 requires a multi-stop navigation response to disclose that the fastest route was chosen for each leg and offer information about alternatives. The stock model sometimes completed the tool action but omitted one of these statements. V14 appends a short neutral disclosure only if all of the following hold in the current user turn: a set_new_navigation proposal exists, an explicitly successful tool result has the same call ID, the visible result contains at least three waypoints, the final event is text rather than another tool call, and the required concepts are absent. This postcondition does

Agent	Base	Hall.	Disamb.	Total
Stock Sonnet 5	9/10	3/10	0/5	12/25
V7-E3	8/10	4/10	3/5	15/25
V14	7/10	4/10	3/5	14/25

Table 1: Strict Pass³ on the same 25-task public panel. V14 is +2 tasks (+8 percentage points) over stock and -1 task (-4 points) versus V7-E3.

not name a task, place, route, or expected answer and makes no LLM call.

2.5 Receipts and Non-Intervention Equivalence

An intervention receipt stores hashes and structured copies of the rejected proposal, visible prefix, advertised tools, derived ledger, corrected proposal, post-validation decision, dispatched call IDs, matched results, and final A2A event. Context identifiers are hashed. On non-intervention paths the same event object and response bytes are forwarded; only final metrics may gain a controller-latency field. This makes both action and non-action inspectable.

3 Validation

3.1 Protocol

We report local public-validation evidence only. Hidden tasks and evaluator state were never inspected. The main comparison reused a frozen list of 25 public tasks (10 Base, 10 Hallucination, and 5 Disambiguation), fixed order and task hashes, and three independent trials per task. A task passes strict Pass³ only when all three trials pass. Completed scored receipts were never rerun. An independent receipt-only parser re-derived every V14 classification; all 25 processes exited normally, task hashes matched, and all 75 trial receipts were present.

Table 1 gives the valid same-basis result. V14 exceeded stock overall but did not exceed V7-E3. Relative to stock, V14 converted three Disambiguation tasks and one Hallucination task, while losing two Base tasks. The four positive task flips did not contain V14 interventions and must be treated as same-model sampling outcomes, not controller-caused gains.

The full panel nevertheless exercises the controller. Across 75 trials, V14 recorded five ID-namespace correction cycles on two Hallucination tasks. Three corrections passed strict post-validation; two were prevented from dispatch after introducing unadvertised parameters. The policy-022 postcondition fired once in each trial of one Base task, with zero fires in the other 72 trials; that task

Mechanism / panel	Fires	Accepted	Strict result
Policy 022 / dev.	3	3	0/3 $\rightarrow$ 3/3
Policy 022 / 25-task	3	3	pass in all arms
ID namespace / 25-task	5	3	no conversion

Table 2: Observed V14 mechanism activity. Two corrected ID proposals were withheld by strict post-validation. “Dev.” is mechanism-development evidence, not sealed generalization.

passed 3/3, as did the stock and V7-E3 arms. These are intervention and safety observations, not an aggregate causal estimate.

3.2 Mechanism Qualification Panel

Before the 25-task V14 run, a three-task development panel had been frozen by a content-blind hash ranking. V7-E3 and the V10 core each passed 2/3 strict tasks. V14 passed all 3/3. On the previously failing Base task, the footer fired in exactly three of three trials and the task moved from 0/3 trials under both comparators to 3/3 under V14; it never fired on the two passing non-Base tasks. This is strong, receipt-bound evidence for the narrow postcondition, but because the mechanism was developed from an observed public failure it is development evidence rather than a sealed generalization estimate.

3.3 Paired Outcome Anatomy

V7-E3 and V14 agreed on 24 of the 25 strict task outcomes. Their sole task-level difference was one Base task, whose trial vector moved from 111 under V7-E3 to 011 under V14 without a V14 intervention. Against stock, V14’s four positive strict flips had vectors 101 $\rightarrow$ 111, 101 $\rightarrow$ 111, 110 $\rightarrow$ 111, and 101 $\rightarrow$ 111; it also lost two Base tasks. A single failed trial changes strict Pass³ from one to zero, so this one-task gap neither validates nor invalidates the architecture.

4 Ablations and Lessons Learned

Why V7-E3 is included. V7-E3 is the most important ablation because it achieved the best raw result: 15/25 versus V14’s 14/25 and stock’s 12/25. Its four positive flips and one Base regression were non-intervention outcomes; its only intervention occurred on a task both arms passed. Thus V7-E3 is a competitive entrant, but its measured +3 tasks cannot be assigned to the controller. Reporting it avoids equating stochastic arm differences with mechanism effects.

Core versus route extensions. V10 isolated the deterministic core from V7-E3’s route-consistency extensions. Offline replay over 492 historical trajectories produced zero fires on 377 known passes and 22 fires on 115 failures, but the extensions had 17 pass-path interference opportunities and were excluded from V14. On the three-task live panel, V10 tied V7-E3 at 2/3; neither satisfied policy 022.

Reasoning scale was not the answer. Increasing reasoning effort on V7-E3 failed the first Base trial and was stopped under a precommitted rule. Earlier Opus experiments reached parity only with materially greater cost. These negatives motivated a precise deterministic postcondition rather than more inference.

Live protocol fidelity mattered. Two postcondition prototypes failed despite passing offline tests. V12 assumed the evaluator’s enriched tool-result representation and mutated no live output. V13 moved the repair to the outer response boundary but still expected a tool name absent from live participant history. V14 instead joins proposals and results through the canonical tool_call_id, which is present in both representations. Preserving these failures was instrumental: the eventual innovation was not the footer text, but binding enforcement to the actual runtime protocol.

5 Integrity and Operational Validation

We used a precommit-and-receipt workflow to reduce adaptive evaluation. Before each panel, candidate source hashes, task hashes, arm order, trial count, budget, and promotion or stop rules were frozen. Candidate failures were retained rather than overwritten. Content-blind task selection used hashes, and no completed scored failure was retried. A second parser consumed only stored receipts and independently reproduced the V14 14/25 total and all category counts.

The public source release contains the participant boundary rather than the private research workspace. Automated checks verify that its controller source contains no Base, Hallucination, or Disambiguation task identifiers. The V14 candidate passed 32 targeted controller and integration tests, including the actual live history shape, call-ID mismatch rejection, single audit emission, single-stop non-activation, compliant-response non-activation, and source-level content-blindness. The local release payload passed a credential scan, served an A2A Agent Card, and completed an A2A message smoke test.

The submitted public OCI manifest is digest-pinned at sha256:d9c06122...540d73bc; anonymous registry inspection reports linux/amd64. The image uses a digest-pinned Python base, runs as UID 1000, and takes provider credentials only from runtime environment variables. The public source is available at <https://github.com/AWF-Z/car-bench-quarry-v14>. These controls do not prove semantic correctness, but they make the reported agent identity reproducible and keep secrets outside the image and source boundary.

The runtime also distinguishes failure modes. If diagnosis itself fails, V14 forwards the stock event rather than inventing a repair. A failed correction call produces a conservative inability response. A corrected proposal that still violates schema or grounding is filtered before dispatch. When an intervention audit path is configured, failure to persist a required receipt is terminal for that intervention. These choices make the

safety/availability trade-off explicit instead of allowing exceptions to silently change behavior.

6 Compute, Cost, and Deployment Assumptions

The evaluation used direct Anthropic inference through LiteLLM, no thinking tokens, no parallel model calls, and sequential task execution. Participant model-call depth is one on normal turns and at most two on correction turns. The deterministic policy-022 path adds no model call. The 25-task V14 campaign observed five correction calls among 408 participant-model calls. Provider sidecars recorded 7,934,543 participant prompt tokens, 93,835 completion tokens, and \$4.3042 participant inference cost. Including the simulated user and evaluator-side calls, the all-provider ledger contained 742 calls and \$6.3492. Timestamp-matched provider receipts assign \$0.1061, or 2.47% of participant spend, to the five correction calls. Their recorded controller latency, which includes the added same-model inference, ranged from 7.94 to 14.54 seconds (median 11.40); summed over all 408 participant calls, these observed correction paths contribute 0.139 seconds per call when amortized. Pass-through overhead was not separately benchmarked. These are local receipt totals, not guaranteed billing or hidden-set estimates; prompt-token accounting can include cached context.

No accelerator, external database, or nonstandard service is required. The model name, API base, thinking mode, and reasoning selector are environment configurable. The container requires outbound HTTPS to the provider and an API key at runtime. We did not benchmark concurrent throughput or claim a general latency improvement. Deterministic work is included in the reported A2A controller_latency_ms; rate-limit behavior therefore depends mainly on the configured model route and the benchmark’s sequential harness.

7 Deployment Implications

V14 suggests three practical patterns beyond the particular automotive tools in CAR-bench. First, a reliability controller should operate on semantic types and observable state transitions, not benchmark labels. “Identifier returned by a POI search,” “navigation is already active,” and “tool result explicitly succeeded” are runtime facts that remain meaningful when prompts and tasks change. This differs from a reward-driven verifier: the controller cannot see whether a trial passed and therefore cannot repeatedly optimize against the metric. Its authority is limited to invariants that an application owner could state before deployment.

Second, sparse intervention is operationally preferable to unconditional multi-agent review when the protected invariants are precise. V14 forwards the stock event when no rule fires, makes at most one extra model call when a guard does fire, and uses no additional inference for policy-022 completion. This keeps the cost surface

explainable: operators can count corrections and deterministic completions separately from normal model usage. The five observed correction calls were only 1.23% of the 408 participant calls in the public panel. The architecture does not claim that every failure is detectable; it instead makes a narrow precision-oriented promise about which failures it is allowed to touch.

Third, receipts turn controller behavior into reviewable production evidence. A digest of the proposal alone is insufficient: diagnosing an incident requires the visible facts, applicable schema, exact rule, corrected candidate, post-validation outcome, and matched tool result. V14 preserves this chain while hashing the context identifier. The same format can support regression replay, rule-level activation monitoring, and rollback decisions without logging hidden evaluation state or teaching the runtime about a task’s expected answer.

The V7-E3 comparison also clarifies entrant selection. A Rank-only decision would favor V7-E3 on our same-task public panel. A mechanism-centered decision has different evidence requirements and can favor V14’s narrower, call-ID-grounded intervention even though its raw score is one task lower. We do not use this distinction to redefine the competition metric; rather, it explains why raw performance and deployable innovation are complementary assessments.

The next engineering step is to compile more policy obligations into typed preconditions and postconditions while preserving a zero-fire corpus of known passes. The next evaluation step is a runtime-supported counterfactual boundary that can snapshot the simulated user, provider sampling state, orchestrator, pending A2A events, and environment. Such a boundary would permit matched continuations from the exact intervention point and turn receipt association into stronger causal estimates.

8 Limitations and Conclusion

The small public panel is not a sealed estimate of hidden performance. V14’s +2 tasks over stock are not intervention-attributable, V7-E3 scored one task higher, and the broader guards converted no strict task in this panel. Full stochastic counterfactual replay was also impossible because the runtime cannot snapshot provider sampling and all orchestrator state.

Within these limits, V14 demonstrates a useful deployment principle: let the model handle open-ended planning, but enforce high-confidence state, schema, and postcondition invariants in a sparse deterministic layer. Call-ID grounding, bounded correction, strict revalidation, and causal receipts make reliability interventions cheaper to inspect and harder to overclaim. For safety-critical tool agents, that auditability is itself a practical innovation.

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